What Layerwise Learning Rates Select: Risk Rebound and Subspace Motion in Compressed Representations

Working manuscript

October 20, 2026

Abstract

Improving a pretraining objective need not improve the information retained for an independent task. We study this mismatch in a shallow convolutional model with squaring, global average pooling, and a linear projection. On periodic Gaussian images, its centered representation is exactly linear in standardized frequency powers. A computer-assisted theorem certifies that actual factorized gradient flow first improves the optimal unregularized population projection risk and subsequently exceeds its initial risk by a fixed amount, while the pretraining loss strictly decreases. An explicit matched-function construction identifies a layerwise learning-rate threshold for reversing the initial risk direction. We distinguish selection within a fixed encoder subspace from changes to that subspace: ordinary linear factorization preserves the encoder row space under an isotropic population objective, whereas the square parameterization need not. Ten paired seeds confirm this distinction in population experiments and substantial square-model subspace motion under complete empirical training; finite-data linear motion also has a reproducible component. Population-based intervention predictions are informative early in empirical training but deteriorate late. CIFAR-10 experiments provide a narrower diagnostic: better supervised rank-8 heads recover most of the gap to a fixed encoder's full linear probe, but unsupervised learning-rate ratios do not establish a stable benefit beyond training speed. The results characterize task-dependent selection and its limits, rather than a universal rule for choosing learning rates.

1 Introduction

Pretraining compresses inputs without access to the downstream task. A representation can therefore improve its training objective while rotating away from a direction that a later task needs. Understanding this possibility requires more than a downstream learning curve: regularized probes, finite labels, changing output scale, and different initial representations can each produce apparent performance changes.

We isolate the effect of the training path. Our model is a single convolutional layer followed by squaring, global average pooling (GAP), and a linear projection. For periodic Gaussian inputs, its centered output is $z = Aq = WBq$, where q contains independent standardized frequency powers. This reduction retains the training geometry: we update the convolutional parameters and projection, rather than replacing their optimization by direct optimization of A. The optimal unregularized population linear risk depends only on the row space of A, so output scaling and probe shrinkage cannot explain a change in this risk.

Our first result is a finite-time construction. On an explicit continuous initialization box, factorized gradient flow reaches two fixed relative-energy sections at which risk has respectively improved and risen above initialization. The loss decreases throughout. A separate matched-function construction gives an exact threshold at which the layerwise update ratio changes the initial risk direction. The threshold is specific to this construction; the fixed-gap theorem is independently certified and does not follow by joining local expansions.

The representation being evaluated matters. A compressed projection may lose task information even when an invertible encoder retains everything. In a genuine encoder bottleneck, $K < C < d$, we therefore measure both spaces. For an isotropic population objective $L(A) = F(AA^\top)$, ordinary linear factorization $A = WV$ changes V only by invertible left multiplication over finite time. Its encoder row space remains fixed even while its projection selects different directions inside it. The square model has a different parameter-induced update and can change the encoder row space itself.

Experiments test these statements progressively. Population experiments use 540 trajectories but ten paired initialization seeds as the statistical units. Complete finite-data training then tests whether empirical optimization reaches similar geometric states. Repeated stochastic streams reveal that linear encoder motion also has a stable finite-data component. Task-conditioned intervention predictions work well early but approach a constant-sign baseline late. Finally, CIFAR-10 distinguishes available encoder information from information selected by an eight-dimensional head. This diagnostic supports the relevance of compression direction without establishing a useful unsupervised intervention rule on

natural images.

The paper contributes a certified fixed-gap risk construction, an explicit matched-function direction threshold, and controlled tests separating encoder-space motion from projection-space selection. Neither the general sensitivity to layerwise initialization nor downstream deterioration itself is claimed as new. We also report the empirical failures that limit transferring the mechanism to stochastic training and natural images.

2 Related Work

Initialization and optimization geometry. Kunin et al. [5] analyze how layer imbalance and learning rates change feature learning, including comparisons with the same initial product. Our scale–learning-rate equivalence belongs to this broader setting. Its role here is to connect the actual parameter metric to an explicit independent-task risk threshold, rather than to establish scale sensitivity for the first time.

Projection heads and feature selection. Gui et al. [2] characterize expansion and shrinkage induced by projection heads with fixed encoders. Xue et al. [8] study layerwise feature weighting and report settings with post-projection early-stopping gains. Torralba and Weiss [7] analyze the same broad convolution–nonlinearity–GAP architecture using GUPA; their Appendix M discusses nonunique decompositions, initialization effects, and recognition performance falling while the loss improves. Our question concerns finite-time population risk along actual joint parameter training, with fixed risk gaps and explicit encoder/projection separation.

Nonlinearity and collapse. Jing et al. [3] analyze two-layer linear contrastive dynamics and dimensional collapse; Tian [6] studies the role of nonlinearity in contrastive dynamics. Chaudhry [1] develops a geometric account of projection heads. The Jacobian Gram matrix governing our effective updates is distinct from a pullback loss Hessian. Moreover, our scalar variance remains positive and our texture experiments retain their numerical ranks: changing a task direction is not, by itself, dimensional collapse. Our contribution is scoped to the stated construction and measured models, without an exhaustive priority claim over all equivalent factorizations.

3 Model, Objective, and Risk

Images and effective coordinates. Choose d independent non-DC, non-self-conjugate Fourier pairs of a real periodic image. Let $x_k^F \sim \mathcal{CN}(0, g_k)$ independently and set

$$q_k = \frac{|x_k^F|^2 - g_k}{g_k}, \qquad \mathbb{E}q = 0, \quad \mathrm{Cov}(q) = I. \tag{1}$$

The network is $x \mapsto W\,\mathrm{GAP}((h * x)^2)$, without biases. With the spatial/Fourier normalization fixed in Appendix A, its centered encoder and projection are

$$r_{\mathrm{enc}} = Bq, \qquad z = Aq, \qquad B_{ck} = g_k |h_{ck}|^2, \quad A = WB. \tag{2}$$

For full-support kernels the Fourier coordinates are free; for local kernels they are coupled by the spatial support constraint. We retain that constraint in training and differentiation.

Pretraining objective. For centered view covariance $\Sigma = ASA^\top$ and view-difference second moment D, we analyze the Gaussian Uniformity Plus Alignment objective (GUPA) [7]

$$\begin{aligned} L(A) = {} & \tfrac{1}{2}\,\mathrm{tr}(ADA^\top) - \tfrac{1}{2}\log\det(I+\Sigma) \\ & - \tfrac{1}{2}\,\mathrm{tr}\left((I+\Sigma)^{-1}\Sigma\right). \end{aligned} \tag{3}$$

The energy features are not Gaussian embeddings; GUPA is the objective studied directly, without invoking exact InfoNCE equivalence. In the isotropic texture setting, independent spectral noise of relative power η gives $S = sI$, $D = 2(s-1)I$, $s = (1+\eta)^2$. For $K = 1$, write $A = a^\top$, $V = \|a\|^2$, and $Q = sV$. Then

$$L(a) = (s-1)V + f(sV), \tag{4}$$

$$f(Q) = -\tfrac{1}{2}\log(1+Q) - \frac{Q}{2(1+Q)}. \tag{5}$$

The theorem remains strictly inside the constraint $Q \le 1$. Empirical experiments use the full matrix objective with within-view sample centering and do not normalize individual images or outputs.

Independent tasks. For $Y = \beta^\top q + \xi$, assume $\mathbb{E}[\xi \mid q] = 0$ and $\mathbb{E}\xi^2 = \sigma^2$. The optimal population linear probe satisfies

$$R_{\mathrm{pop}}(A) = \sigma^2 + \|\beta\|^2 - \beta^\top P_A \beta, \tag{6}$$

$$P_A = A^\top (AA^\top)^\dagger A. \tag{7}$$

We denote excess risk by $\mathcal{R} = R_{\mathrm{pop}} - \sigma^2$. Equation (6) is evaluated by an SVD row-space basis, with singular values and numerical ranks recorded. It is invariant to invertible changes of representation coordinates. For $K = 1$, $d = 2$, and $\beta = e_1$, $\mathcal{R} = a_2^2/(a_1^2 + a_2^2)$. The encoder has the same risk formula with B replacing A; an invertible B has risk σ^2 regardless of the head.

For complete orthonormal coordinate tasks, $\sum_j \mathcal{R}_{e_j}(A) = d - \mathrm{rank}(A)$. Thus fixed-rank improvements and losses represent a redistribution of a fixed total budget, not necessarily a decrease in dimension. The substantive question is which directions the training geometry selects.

4 Certified Risk Rebound and Update Geometry

4.1 A finite-time fixed-gap construction

Take $C = d = 2$, $K = 1$, $\beta = e_1$, and unit parameter learning rates. Define the positive initialization coordinates

$$x = (w_1, w_2, B_{11}, B_{12}, B_{21}, B_{22}, g_1, g_2) \tag{8}$$

and the rational center

$$\bar{x} = (1/25, 9/25, 8/25, 4/125, 4/125, 8/125, 1, 2). \tag{9}$$

Each B_{ck} is realized by $|h_{ck}|^2 = B_{ck}/g_k$. Spectral powers g are fixed along each trajectory, but vary across the initialization box. Set $\ell = \log(V/V_0)$ separately for each trajectory.

Theorem 1 (Certified fixed risk gaps)**.** *Let $\delta = 10^{-5}$ and let $x_i(0) \in [(1-\delta)\bar{x}_i, (1+\delta)\bar{x}_i]$ independently. For every $\eta \in [0, 1/5]$, Euclidean parameter gradient flow for the isotropic GUPA objective reaches $\ell_1 = 1.7$ and $\ell_2 = 6$ in finite physical time. Throughout this interval the parameters and vector-field denominators remain positive and $Q < 0.689 < 1$. Moreover,*

$$\mathcal{R}(1.7) \le \mathcal{R}(0) - 0.04, \tag{10}$$

$$\mathcal{R}(6) \ge \mathcal{R}(0) + 0.15, \tag{11}$$

and hence $\mathcal{R}(6) - \mathcal{R}(1.7) \ge 0.19$. The pretraining loss strictly decreases between these times and initialization.

The relative half-width is 0.001%, not 1%. This is a continuous-box existence result, not a high-probability random-initialization theorem. The guarantee concerns two sections and does not assert a unique risk minimum or monotonicity between sections. It does not assert encoder deterioration: whenever the two-dimensional encoder is invertible, its optimal encoder risk is at the noise floor. The theorem concerns the compressed projection.

Proof structure. Write $\gamma(Q) = (Q+2)/(2(Q+1)^2)$ and $\Gamma_s(Q) = s\gamma(Q) - (s-1)$. A positive time change $d\zeta/dt = 2\Gamma_s(Q)$ gives

$$w_c' = \sum_k B_{ck} a_k, \qquad B_{ck}' = 4 g_k B_{ck} w_c a_k. \tag{12}$$

For $K_B = B^\top B + \mathrm{diag}(4g_k \sum_c B_{ck} w_c^2)$ and $H = a^\top K_B a$, $d\ell/d\zeta = 2H/V > 0$. The resulting rational vector field is integrated with outward-rounded fixed-point intervals. A Taylor enclosure certifies the center trajectory, and a logarithmic-coordinate perturbation bound encloses every initialization in the box. Strict positivity, the energy bound, and $\Gamma_s \ge 0.1$ ensure finite physical reachability. Appendix B supplies the enclosure argument and exact reproduction settings.

The stored certificate is stronger than the displayed conservative theorem: it gives improvement at least 0.09122, excess over initialization at least 0.33664, and rebound at least 0.42790. These are certified bounds, not sampled means.

4.2 Matched functions can move in opposite task directions

Proposition 2 (Scale and learning-rate equivalence)**.** *For a loss depending on $A = WB(h)$ with $B(ch) = c^2 B(h)$, scaling $h \mapsto ch$ and $W \mapsto c^{-2}W$ preserves the initial function. Euclidean gradient training at this scaled initialization is equivalent, in the unscaled coordinates, to learning rates*

$$(\eta_h, \eta_W) \mapsto (c^{-2}\eta_h, c^4 \eta_W). \tag{13}$$

This holds step by step for ordinary gradient descent. The relative rate acquires the factor c^6.

Proof. In coordinates $h_c = c\bar{h}$, $W_c = c^{-2}\bar{W}$, the gradients scale as $\nabla_{h_c} L = c^{-1}\nabla_{\bar{h}} L$ and $\nabla_{W_c} L = c^2 \nabla_{\bar{W}} L$. Substituting into the update gives the stated factors. □

An explicit direction test uses a different initialization from Theorem 1:

$$B_0 = \begin{pmatrix} 2 & 1/5 \\ 1/5 & 2/5 \end{pmatrix}, \quad w_0 = \begin{pmatrix} 1/10 \\ 9/10 \end{pmatrix}, \quad g = (1, 2). \tag{14}$$

All scaled versions have $a_0 = (19/50, 19/50)^\top$. For $S = I, D = 0$, direct differentiation yields

$$\left.\frac{d\mathcal{R}}{d\ell}\right|_0 = \frac{235 - 480c^6}{834 + 1300c^6}. \tag{15}$$

The sign reverses at $c_* = (47/96)^{1/6} \simeq 0.887778$, or relative learning rate $r_* = 47/96$ in the unscaled coordinates. Opposite derivatives at the same initial representation and equal-energy section cannot be removed by a positive time change. This threshold is specific to the displayed construction, not a universal recommended ratio. The fixed-gap initialization is a separate scaled-down state and is certified independently.

4.3 Which space changes?

Proposition 3 (Linear population row-space conservation)**.** *Suppose $A = WV$, $L(A) = F(AA^\top)$, and $G = \nabla_A L = H(A)A$, with H continuous along a finite, bounded parameter trajectory. Then ordinary linear gradient flow preserves* $\mathrm{row}\, V(t) = \mathrm{row}\, V(0)$*. Every task's optimal population encoder risk is constant.*

Proof. The front-layer update is

$$\dot{V} = -\eta_V W^\top H(A) W V = C(t) V. \tag{16}$$

Its solution is $V(t) = U(t)V(0)$, where $U' = C(t)U$, $U(0) = I$, and $\det U(t) = \exp(\int_0^t \mathrm{tr}\, C(s) ds) \neq 0$. Left multiplication preserves row space; Equation (6) completes the proof. □

This statement covers matrix outputs, not just $K = 1$. The front-layer contribution to $\dot{A}$ is also a left multiplication of A, so its first-order contribution to row-space risk is zero. At a fixed state, changing a positive rate ratio cannot reverse this linear model's instantaneous population risk sign; finite horizons can still differ through later states.

For the square model, the actual full-support parameter update instead gives

$$\dot{B}_{ck} = -4\eta_h g_k B_{ck}(W^\top G)_{ck}. \tag{17}$$

There is generally no common left multiplier of B. In a bottleneck $C < d$, its encoder row space can therefore move. This does not imply that movement is always harmful or necessary for projection risk rebound. Linear factorization itself can produce substantial projection motion within its conserved encoder space.

4.4 Task-conditioned local interventions

For a full-row-rank representation matrix X, let $v = (XX^\top)^{-1}X\beta$ and $e = (I - P_X)\beta$. Differentiation of the least-squares residual gives

$$\dot{\mathcal{R}} = -2v^\top \dot{X} e. \tag{18}$$

Separating parameter groups yields $\dot{\mathcal{R}} = \eta_h c_h + \eta_W c_W$, where $c_h = -\langle \nabla_h R, \nabla_h L\rangle$ and likewise for W. Opposite signs permit a positive critical ratio $r_* = -c_h/c_W$; equal nonzero signs exclude immediate reversal by positive rates. We calculate coefficients from effective maps and update geometry before executing branches. The rule uses known task direction β and is an oracle mechanism test, not a label-free controller or a guarantee of long-horizon benefit.

5 Controlled Texture Experiments

5.1 Protocol and numerical checks

We generate 32×32 periodic textures using four directions, two radial bands per direction, and two independent conjugate frequency pairs per group, for $d = 16$ independent power coordinates. The initial spectrum is flat. Images remain floating point without per-image normalization or clipping. Pretraining uses no downstream labels. Ten predefined unit-variance tasks comprise two individual power coordinates and eight direction/band energy sums; all are evaluated on the same checkpoints.

The main bottleneck has $C = 8$, $K = 1$; extensions use $K = 2$ or 11×11 periodic kernels. Parameters have no bias, batch normalization, weight decay, momentum, or output normalization. We compare $r = \eta_W/\eta_h \in \{0.1, 1, 10\}$. Each seed shares initial effective representations and paired data/augmentation streams across ratios. Controls are direct effective-map optimization, ordinary linear factorization on the same q, and frozen layers. A linear control on raw pixels would change task expressivity and is not used.

Table 1: Population subspace displacement at $\ell = 3$, independently calibrated $r = 1$, ten paired seeds. Linear encoder conservation coexists with projection motion.

	Encoder D_B		Projection D_A	
Setting	Square	Linear	Square	Linear
$K = 1$, full	0.3064	$1.15\,10^{-14}$	0.5161	0.6087
$K = 2$, full	0.4499	$1.23\,10^{-14}$	0.5597	0.5319
$K = 1$, local	0.3786	$1.26\,10^{-14}$	0.5572	0.6196

We retain two calibration protocols. The matched-parameter protocol starts from the same B_0, W_0. Independent calibration instead uses each model's parameter Jacobian Gram traces, without task labels, to equalize its two active blocks and normalize initial total mobility. Scalar factor rescaling preserves A_0 and encoder row space. Equal traces do not imply equal update directions. Development seeds 0 and 1 are separate from ten confirmation seeds 2–11. Population protocols were locked before confirmation; settings reuse the same ten seeds and do not supply independent repetitions.

Spatial convolution, FFT evaluation, and effective-map gradients are checked independently. Centered spatial output agrees with Aq to approximately 9.71×10^{-17}, and the spatial/FFT gradient discrepancy is approximately 6.94×10^{-18}. The real-parameter fixed-section error at $\ell = 6$ decreases from approximately 4.02×10^{-4} to 2.01×10^{-4} when the step is halved. An independent empirical gradient check has approximately 1.98% relative error, reflecting moment estimation rather than machine-precision equivalence. No finite-sample trajectory is treated as a pointwise theorem reproduction.

5.2 Population subspace selection

The confirmation suite contains three architectures, two calibrations, three models, three ratios, and ten paired seeds: 540 population trajectories. Its primary comparison is the common section $\ell = 3$. Let $D_X = \left\|P_{X(t)} - P_{X(0)}\right\|_F$. Table 1 reports independently calibrated $r = 1$ means.

Every observed square-model seed has resolvable encoder motion, whereas linear encoder displacement remains at rounding scale. Frozen-front displacement is zero. All observed encoder and projection ranks remain 8 and K. Changing r also changes square-model displacement: for $K = 1$ full support, its mean is 0.7161, 0.3064, and 0.0485 for the three ratios. These are geometric changes, not uniformly worse task risks. Direct optimization of a scalar effective map under the radial population objective preserves its direction, while ordinary linear factorization can change projection risk. Rebound is

Table 2: Encoder displacement after complete empirical training, evaluated at common $\ell = 1.5$; ten-seed means. Frozen-front values are zero.

Kernel	Model	$r = 0.1$	$r = 1$	$r = 10$
Full	Square	0.41312	0.13692	0.01843
Full	Linear	0.02336	0.00666	0.00089
Local	Square	0.44250	0.17740	0.02626
Local	Linear	0.02851	0.00920	0.00123

therefore not unique to the square architecture.

5.3 Complete empirical training and reproducible drift

We train from initialization on 32,768 unlabeled images, batch size 256, for 4,000 updates. Full-support and local-kernel suites each contain 90 trajectories: three models, three ratios, and ten seeds. All reach the prespecified common section $\ell = 1.5$. Forty additional trajectories provide three independent SGD streams per seed for the square and linear models at the repeated condition. Table 2 reports encoder displacement.

Empirical square-model geometry remains close to the corresponding population state: at $r = 1$, $\ell = 1.5$, the empirical/population encoder projector distance averages 0.00713 for full support and 0.00850 for local kernels, compared with movements of 0.13692 and 0.17740 from initialization. This agreement concerns this model family, dataset size, and training budget.

Linear encoder movement is nonzero and reproducible. Conditional on initialization and the finite dataset, the mean pairwise cosine between terminal projector changes from independent streams is 0.9810 for the square model and 0.8683 for the linear model. These are within-seed agreements, not evidence for a universal direction across initializations. Finite datasets need not have isotropic power covariance, so their conditional expected gradient need not satisfy Proposition 3. Exploratory conditional-moment calculations on two development states support this explanation. Nonlinear sample-moment estimation and finite update size may also contribute; the diagnostic is not an independently confirmed decomposition of all drift.

5.4 Intervention prediction has a training-stage boundary

The final test uses checkpoints at updates 500, 2,000, and 3,500 from complete empirical training. Each of 90 model/seed/checkpoint states has 32 independent future batch/augmentation streams. At the midpoint, eight streams continue for 8 and 64 updates. Population geometry predicts task-conditioned risk changes before drawing future batches; actual branches use empirical GUPA

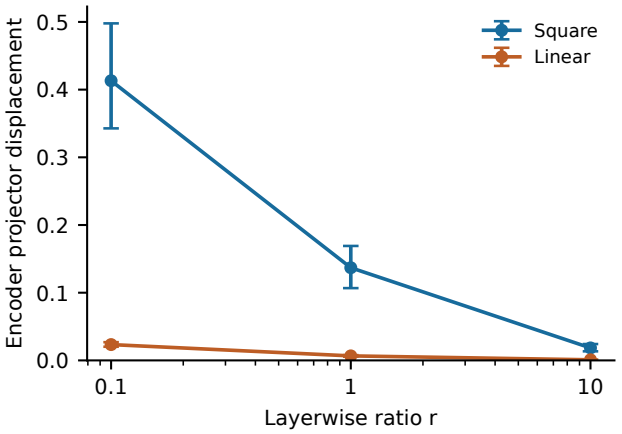


Figure 1: Full-support empirical encoder displacement at common $\ell = 1.5$. Error bars are descriptive paired-seed bootstrap intervals from the saved summary. Square-model motion is larger, but finite-data linear motion is nonzero.

updates. Seed averages, rather than branches or tasks, determine summaries.

For the square model, single-step sign agreement is 91.8%, 57.0%, and 52.6% at the three stages; constant-decrease baselines are 55.6%, 47.7%, and 50.8%. At the midpoint the sign of the repeated-batch mean agrees 92.3% of the time even though individual batches agree only 57.0%. Late agreement is close to baseline for both summaries. This distinguishes prediction of an average direction from reliable control of a single stochastic update.

Longer midpoint windows raise sign agreement to 69.8% and 71.0%, but amplitude mean absolute error rises from approximately 1.2×10^{-5} for one update to 6.7×10^{-4} for 64 updates. Noise averaging can improve signs while the first-order magnitude approximation worsens. For the prespecified primary task, positive population thresholds occur in 5/10 early states and only 1/10 at each later stage. Threshold errors are reported only where comparable positive thresholds exist; we do not force every state to admit a reversal. The intervention remains an oracle mechanism experiment using known β.

6 CIFAR-10: A Compression Diagnostic

Natural-image experiments test the scope of the mechanism rather than reproduce the two-coordinate threshold. We use standard CIFAR-10 [4], grayscale input, 64 11×11 valid convolutional filters, squaring, GAP, and an eight-dimensional head. Training starts from scratch, without downstream labels. The training split is 45,000/5,000, and the standard 10,000-image test set is reserved for the fixed final-budget comparison. Augmentations are

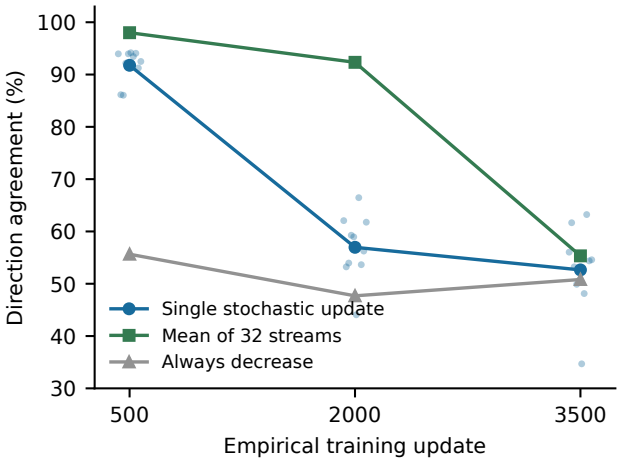


Figure 2: Square-model one-update direction predictions at empirical checkpoints. Markers average ten seed-level summaries; thin points show seeds. The distinction between individual stochastic updates and replicate means disappears late.

reflected crops, horizontal flips, contrast changes, and additive noise. Classification uses unregularized affine OLS on one-hot labels, evaluating encoder and projection separately. Natural-image accuracy is not the theorem's population square risk.

6.1 Layerwise rates do not establish a speed-independent gain

Five paired seeds and three ratios run for 4,000 updates at batch size 256, approximately 22.8 training-set passes. Encoder test accuracy is 36.48%, 36.45%, and 36.52% for $r = 0.1, 1, 10$; projection accuracy is 23.39%, 23.50%, and 23.81%. The fixed-budget projection difference is approximately 0.43 percentage points before rounding the group means. At common calibration energy $\ell = 2$, however, the corresponding validation difference is approximately $+0.11$ points with a five-seed descriptive bootstrap interval $[-0.42, +0.54]$. Other common-energy intervals also cross zero. Energy is measured on a fixed unlabeled calibration set and its first crossing is checked every ten updates; it is a proxy with recorded overshoot, not an exact population progress variable. The results do not establish either a stable speed-independent benefit or its absence.

6.2 More capacity mostly improves the initial features

Development diagnostics use seeds 0, 1, and 12, three architectures, and two supervised references, for 33 runs of 4,000 updates. These runs use validation labels only for evaluation, never the test set. Table 3 separates initial performance from learned improvement.

Table 3: Validation OLS accuracy (%) for capacity diagnostics, $r = 1$; means over three development seeds. Each cell shows initialization $\rightarrow$ 4,000 updates.

Architecture	Encoder	Projection
$C = 64, K = 8$	$36.31 \rightarrow 36.04$	$23.49 \rightarrow 23.71$
$C = 256, K = 8$	$43.04 \rightarrow 43.04$	$23.66 \rightarrow 23.65$
$C = 64, K = 32$	$36.31 \rightarrow 36.23$	$32.05 \rightarrow 32.14$

Table 4: Same frozen encoder, different heads. Validation accuracy (%), three paired development seeds per source state.

Readout	Initial	Trained
Full encoder OLS	36.31	36.04
Original head + OLS	23.49	23.71
Rank-8 least squares	35.75	35.59
CE-trained head + OLS	30.89	28.49
CE-trained classifier	30.83	27.73

Width and projection dimension improve absolute readability, but their gains largely exist at random initialization. Supervised training of the original model reaches encoder OLS accuracy 40.07% through the encoder path and projection OLS accuracy 38.69% through the joint projection path. Thus 24% is not an expressivity ceiling for eight-dimensional outputs. The supervised references use Adam and prescribed gradient clipping, unlike unsupervised SGD, and are attainable references rather than matched optimization or capacity upper bounds.

The wide model has a small exploratory ratio signal: $r = 10$ minus $r = 0.1$ projection accuracy is positive for all three development seeds at $\ell = 1$ and 2, with means near $+0.20$ and $+0.19$ points. Multiple inspected sections, few development seeds, and inconsistent MSE signs at some sections prevent a stable intervention claim. Some $K = 32$ energy sections are missing; we retain those missing cases rather than treating available pairs as a complete experiment.

6.3 A fixed encoder admits a much better eight-dimensional head

We freeze the original $r = 1$ encoder at initialization and update 4,000 for each development seed: six source states, with only three independent seeds. Each state compares the original head, full encoder OLS, an optimal rank-8 affine least-squares fit, and a supervised head trained with cross-entropy (CE). The rank-8 solution uses training labels only. Its affine intercept is absorbed into the final decoder, leaving the projection itself linear. CE training updates only the projection and classifier; kernels are checked for exact equality before and after training.

For the trained encoder, rank-8 fitting improves accuracy over the original head by approximately 11.88 points,

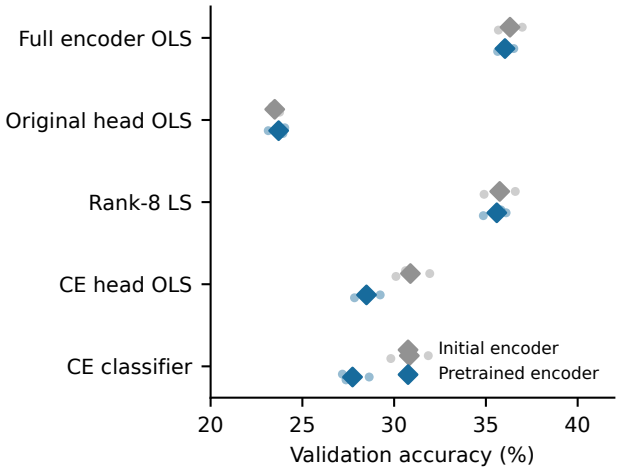


Figure 3: Validation accuracy for different readouts of the same fixed encoder. Points are paired development seeds. The large source-head gap exists both before and after pretraining. Rank-8 fitting is supervised.

remaining only 0.45 points below full encoder OLS. Its validation one-hot MSE is 0.08147432, compared with 0.08631537 for the original head and 0.08141745 for the full encoder. It removes 98.84% of this particular MSE gap; this is not a percentage of all semantic information. A better compressed choice exists without any encoder change.

Crucially, almost the same gap is already present at initialization. These data do not show that pretraining created a large new semantic loss. The CE-trained head improves over the source head but falls short of rank-8 fitting even on training accuracy (28.71% versus 36.59%). The gap is therefore not solely a validation overfitting effect, but CE versus square loss, augmented versus clean inputs, scaling, clipping, and budget are confounded; factorized optimization alone is not identified as its cause.

Ten-class centered one-hot targets have at most nine independent directions, so near-recovery by eight coordinates has a direct algebraic context. Reduced-rank fitting is globally optimal for training MSE, not validation accuracy. It establishes available task-dependent compression, not a label-free way to find it or an architecture-specific phenomenon.

7 Discussion and Limitations

Layerwise rates affect the metric through which pretraining selects directions. Our certified construction shows that this can produce a fixed rise in an independent task's optimal population projection risk while the loss decreases. The controlled high-dimensional experiments then distinguish two effects: reselecting directions within a fixed encoder space and changing the encoder space itself. Neither effect should be inferred from a projection curve alone.

The population distinction is conditional. Finite data break exact isotropy, empirical nonlinear moment objectives need not have unbiased gradients, and finite steps contribute additional changes. Linear encoder movement under complete empirical training has a reproducible component; describing it as only zero-mean stochastic noise would be incorrect. Likewise, early success of population-based risk predictions does not imply a reliable late-stage stochastic intervention. We report both failures explicitly.

The theorem is computer-assisted rather than a proof-assistant formalization. Its continuous initialization box is narrow, and the high-dimensional random-seed experiments do not enlarge its certified domain. Its radial example is not contrastive-specific: smooth decreasing radial objectives share the same energy-parameterized path over intervals where their scalar speed remains positive. Full-support coordinates are permutation-equivariant and do not by themselves explain a universal low-to-high frequency order. Local-kernel experiments add a support constraint, not a new locality theorem.

Natural-image evidence is deliberately limited. The shallow grayscale architecture, finite training budget, and small seed counts do not support general claims about visual backbones. Fixed-encoder rank-8 fitting shows that head selection can discard useful class directions, but it uses labels and the gap already exists at initialization. Covariance effective-rank concentration is also not equivalent to loss of all linearly readable information. The current evidence supports a controlled account of task-dependent representation selection and a measurable boundary on its predictability.

8 Conclusion

Actual factorized training can improve its pretraining objective while first improving and then worsening the optimal population risk of a compressed representation. We certify this behavior on an explicit initialization box and identify a matched-function learning-rate threshold for its initial direction. Population and empirical experiments separate projection selection from encoder subspace motion, while exposing finite-data drift and late-stage prediction failure. CIFAR-10 diagnoses substantial head-selection gaps without demonstrating a stable unsupervised ratio benefit. These distinctions make clear which parts of the phenomenon are explained, which depend on idealized assumptions, and which remain unresolved.

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A Spatial Metric and Effective Updates

For image side length n, use orthonormal real cosine/sine basis functions $\sqrt{2/n^2}\cos(2\pi k\cdot x/n)$ and the corresponding sine functions. Circular correlation is divided by $\sqrt{n^2/2}$. The physical spatial kernel then has Euclidean coordinates $h_{ck,\cos}, h_{ck,\sin}$, and

$$B_{ck} = g_k(h_{ck,\cos}^2 + h_{ck,\sin}^2). \tag{19}$$

This convention counts each independent conjugate pair once. The effective centered map is obtained by subtracting its distributional mean; empirical GUPA uses sample centering and probes include an intercept. Differentiating physical kernels gives Equation (17), including its factor of four. Fourier SGD with a different normalization is not automatically the same algorithm.

For a local kernel let $J_c = D_{h_c}B_{c:}$ and $T_c = J_cJ_c^\top$. Then, in column-vector notation,

$$\dot{B}_{c:}^\top = -\eta_h T_c(W^\top G)_{c:}^\top. \tag{20}$$

For full support $T_c = \mathrm{diag}(4g_kB_{ck})$; local support introduces cross-frequency terms. These terms are retained in the experiments. Effective updates are parameter Jacobian Gram updates, not arbitrary effective-map optimization.

B Details of the Certified Fixed-Gap Theorem

B.1 Energy-parameterized rational dynamics

Let $a = B^\top w$, $V = a^\top a$, $K_B = B^\top B + \mathrm{diag}(4g_k\sum_c B_{ck}w_c^2)$, and $H = a^\top K_B a$. In effective time, Equation (12) gives $a' = K_Ba$ and $V' = 2H$. Positivity implies $H > 0$. In $\ell = \log(V/V_0)$ coordinates, the eight-dimensional vector field is

$$F_{w_c}(x) = \frac{V}{2H}\sum_k B_{ck}a_k, \tag{21}$$

$$F_{B_{ck}}(x) = \frac{V}{2H}4g_kB_{ck}w_ca_k, \tag{22}$$

$$F_{g_k}(x) = 0. \tag{23}$$

Along exact solutions $dV/d\ell = V$, hence $V(\ell) = V_0e^\ell$. Although each initialization has its own V_0, the rational field contains no external V_0 parameter. This permits a common-section comparison across the box.

B.2 Outward-rounded trajectory enclosures

The certificate uses step $h = 1/500$, 3,000 steps, and fixed-point endpoints with denominator 2^{180}. The two sections are exactly steps 850 and 3,000. All proof decisions use integer outward rounding; floating point is used only for readable progress output. Any denominator containing zero causes failure.

For each center-trajectory endpoint enclosure X_n, the verifier constructs a positive trajectory tube T_n and checks strict Picard inclusion in the six dynamic coordinates:

$$X_n + [0,h]F(T_n) \subset \mathrm{int}(T_n). \tag{24}$$

The two static coordinates retain their input intervals. A first-exit argument establishes existence and containment throughout the step. Truncated power-series arithmetic computes coefficients recursively,

$$c_0 = x, \qquad c_{j+1} = \frac{[z^j]F(\sum_{k=0}^j c_kz^k)}{j+1}. \tag{25}$$

The step endpoint satisfies the interval Taylor inclusion

$$x(\ell_n + h) \in \sum_{j=0}^{7} c_j(X_n)h^j + c_8(T_n)h^8. \tag{26}$$

Thus the enclosure includes the eighth-order remainder, all arithmetic rounding, and the entire within-step trajectory. Checking only endpoints would not suffice.

B.3 Enclosing the continuous initialization box

Directly propagating an initial box can inflate intervals. Instead, enclose the center and control neighboring solutions in logarithmic coordinates $y = \log x$. Define $f_i(y) = F_i(e^y)e^{-y_i}$ and $J = D_yf$. For the initial relative half-width $\delta = 10^{-5}$,

$$\|y(0) - \bar{y}(0)\|_2 \le \sqrt{8}\frac{\delta}{1-\delta} < \frac{3}{99999} = E_0. \tag{27}$$

For each step the verifier certifies

$$\lambda_{\max}\big((J + J^\top)/2\big) < m_n \tag{28}$$

throughout the inflated trajectory tube. It searches $m_n \in \{1/20, 2/20, \ldots, 80/20\}$ and sets, with upward rounding,

$$r_n = \frac{E_n}{1 - m_nh}, \qquad E_{n+1} = r_n < 1/2. \tag{29}$$

Since $e^{m_nh} < 1/(1-m_nh)$, this bounds the whole step. The required tube is the componentwise product

$$\widehat{T}_n = T_n[1 - r_n, (1-r_n)^{-1}]. \tag{30}$$

It contains exponentials of the log-error ball and the log-coordinate segments joining neighboring solutions. The derivative is evaluated without approximate logarithms:

$$J_{ij} = \frac{x_j}{x_i}\frac{\partial F_i}{\partial x_j} - \mathbf{1}_{i=j}\frac{F_i}{x_i}. \tag{31}$$

The static g rows are zero, but their influence on other rows remains included.

To prove Equation (28), let $\mathscr{S}$ enclose the symmetric part of J, let M be its symmetric fixed-point midpoint matrix, and let e_n bound the maximum absolute row sum of interval deviations from M. Every real $S \in \mathscr{S}$ obeys $\|S - M\|_2 \le e_n$. The verifier checks

$$(m_n - e_n)I - M \succ 0 \tag{32}$$

by outward-rounded unpivoted $LDL^\top$ recursion with all eight pivots strictly positive. Failure of every candidate m_n aborts verification. Integrating the symmetric-Jacobian bound along segments and using a first-exit argument closes the error tube. No approximate eigenvalue is accepted as a proof decision.

B.4 Risk bounds, constraints, and physical time

At a section, inflate X_n by $[1 - E_n, (1 - E_n)^{-1}]$. For $a_1 \in [A_-, A_+]$, $a_2 \in [C_-, C_+]$ with positive endpoints, use

$$\mathcal{R} \in \left[\frac{C_-^2}{A_+^2 + C_-^2}, \frac{C_+^2}{A_-^2 + C_+^2}\right]. \tag{33}$$

Subtracting section and initialization intervals gives uniform bounds for each initialization, conservatively ignoring their correlations. Readable outward bounds are

$$0.49997 \le \mathcal{R}(0) \le 0.50003, \tag{34}$$

$$0.40844 \le \mathcal{R}(1.7) \le 0.40876, \tag{35}$$

$$0.83666 \le \mathcal{R}(6) \le 0.83883. \tag{36}$$

The stronger gap constants quoted in the main text use the unrounded binary-rational endpoints, not subtraction of these rounded displays.

The energy bound holds continuously, using $\bar{V}_0 = 0.0011829248$ and an exact rational series proof of $e^6 < 404$:

$$Q(\ell) \le \frac{36}{25}(1+\delta)^4 \bar{V}_0 e^6 < 0.689. \tag{37}$$

For $1 \le s \le 36/25$ and $Q \le 1$, $\Gamma_s(Q) \ge 0.1$. Therefore

$$\frac{d\ell}{dt} = 4\Gamma_s(Q)\frac{H}{V} > 0, \qquad \frac{dL}{d\ell} = -\Gamma_s(sV)V < 0. \tag{38}$$

The compact positive tube has a uniform lower bound on H and an upper bound on V, implying finite $dt/d\ell$ and finite physical reachability. This completes the computer-assisted proof of Theorem 1, conditional on the explicitly checkable arithmetic and inclusion tests.

A matched-function subfamily. Fix $g = (1, 2)$ at the center and scale $B_0 = c^2\bar{B}$, $w_0 = c^{-2}\bar{w}$ with $c \in [1 - 1/400000, 1 + 1/400000]$. Rational endpoint checks give $c^2, c^{-2} \in [1 - \delta, 1 + \delta]$. This whole family lies in the certified box and has the same $a_0 = (0.02432, 0.02432)^\top$ and $\mathcal{R}(0) = 1/2$. The full box itself does not match a single initial representation.

Reproduction. From the repository root, run `python -B` on `experiments/code/theory/verify.py` and `experiments/code/theory/check_arithmetic.py`, with assertions enabled. The verifier reconstructs the certificate from rational initialization without reading the stored certificate. The independent Fraction-based arithmetic check does not replace the ODE enclosure. The proof is not formalized in Lean or Coq.

C Derivation of the Direction Threshold

Let $P = B^\top B$, $F_B = \mathrm{diag}(4g_k \sum_c B_{ck} w_c^2)$, and $K_B = P + F_B$. For the matched-function initialization in Section 4,

$$P_0 = \begin{pmatrix} 101/25 & 12/25 \\ 12/25 & 1/5 \end{pmatrix}, \quad F_0 = \mathrm{diag}(91/125, 326/125). \tag{39}$$

Scaling gives $K_B = c^4 P_0 + c^{-2} F_0$. With $a_* = 19/50$ and $a_\perp = (-a_2, a_1)^\top$,

$$N_0 = a_\perp^\top K_B a = a_*^2 c^{-2}(47 - 96c^6)/25, \tag{40}$$

$$a_0^\top K_B a_0 = a_*^2\big((26/5)c^4 + (417/125)c^{-2}\big). \tag{41}$$

At equal positive coordinates, the derivative of $\mathcal{R} = \sin^2\vartheta$ with respect to angle is one. Thus $d\mathcal{R}/d\ell = N_0/(2a_0^\top K_B a_0)$, which simplifies to Equation (15). The threshold changes only the initial sign. The main fixed-gap theorem does not infer a long-time trajectory from this local calculation.

More generally, any smooth radial objective $\Phi(V)$ with $-\Phi'(V)$ bounded below by a positive constant on the trajectory interval shares Equation (12) after a positive time change. The energy-parameterized paths are consequently independent of the particular radial loss on that interval. Anisotropic objectives do not generally have this scalar-factor property.

D Finite-Label Probes and Reduced-Rank Fitting

Equation (6) follows by minimizing $\sigma^2 + \left\|\beta - A^\top v\right\|^2$ over v. An invertible feature coordinate transformation preserves the feasible prediction space. For a full-column-rank finite design and an invertible encoder, unregularized affine OLS is similarly unchanged in exact arithmetic. Fewer labels alone do not prove encoder deterioration; minimum-norm conventions in underdetermined settings, regularization, or numerical truncation require separate analysis.

For the frozen-encoder experiment, center and invertibly column-standardize the training feature matrix to obtain X, and center training one-hot targets to obtain

Y. Write the numerical-rank SVD as $X = USV^\top$. Orthogonal decomposition gives

$$\|Y - XM\|_F^2 = \left\|(I - UU^\top)Y\right\|_F^2 + \left\|U^\top Y - SV^\top M\right\|_F^2. \tag{42}$$

The best rank-K approximation of $U^\top Y$ is its truncated SVD. Mapping back through VS^{-1} gives a rank-K linear feature projection and decoder; centering is absorbed in decoder bias. This globally minimizes training square error over rank-K affine predictors, without using validation labels. It is not globally optimal for classification accuracy or for the CE training objective. The source implementation additionally refits an OLS decoder to the resulting projection, enabling an independent consistency check.

E Experimental Scope and Reporting

Population suites contain 540 trajectories, complete empirical suites 180 primary trajectories, and random-stream repeats 40 additional trajectories. These counts are computational workloads. The texture inferential unit remains ten paired seeds. Repeated checkpoints, ten tasks, and future batch streams are correlated measurements. CIFAR fixed-budget comparison uses five paired seeds; capacity and frozen-head diagnostics use three development seeds, not new confirmation samples.

The theoretical and population sections use exact known-distribution risks. Texture empirical training still evaluates those risks analytically rather than fitting them on test labels. Separate pilot finite-label OLS uses nested label counts 256, 1,024, and 8,192; it is not part of the proof. Natural-image OLS fits on the 45,000-image training split. Only the fixed-budget five-seed comparison uses the standard test set; the later capacity and frozen-head diagnostics use validation only.

For interventions, predicted changes with absolute magnitude at most 10^{-9} are treated as numerically neutral and excluded from sign summaries. Positive-threshold existence and conditional threshold errors are reported separately. Early and late states cannot be pooled into a single reliability claim. Population-trained checkpoint branching is preliminary evidence; the final intervention table uses states reached by complete empirical training.

The CIFAR supervised references use Adam, front/head learning rate 3×10^{-4}, classifier learning rate 10^{-3}, and prescribed global gradient clipping at 1. Fixed initial feature standardization is used only in their classifier path. These choices preserve the represented function family but change optimization relative to unsupervised SGD. The local supervised stability tests triggered clipping on approximately 98–99% of updates; those local counts do not independently certify clipping rates for server runs. Current server capacity and frozen-head evidence was imported as summaries; raw server checkpoints and optimization logs were not re-audited locally.